IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

EEE PART II: MEng, BEng and ACGI

MATHEMATICS 2B (E-STREAM AND I-STREAM)

Friday, 22 May 10:00 am

Time allowed: 1:30 hours

There are TWO questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : B. Clerckx
Second Marker(s) : O. Sydoruk

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THE QUESTIONS

[25]

1. a) Consider the non-negative random variable X characterized by

$$P(X \leq x) = \begin{cases} 1 - e^{-x^2}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

i) Compute the probability $P\left(X^2 \geq 2a | X^2 > a\right)$ (with $a > 0$). Provide your reasoning.

[4]

ii) Compute the expectation of X, i.e. $E(X)$. Provide your reasoning.

[4]

iii) Compute the probability density function of X^2. Provide your reasoning.

[3]

iv) Compute the variance of X, i.e. $\mathrm{Var}(X)$. Provide your reasoning.

[3]

b) Consider the random sample $X_1, X_2, \ldots, X_n$ from a Normal distribution $N\left(\mu, \sigma^2\right)$.

i) Determine the maximum likelihood estimator of μ and σ^2 (denoted as $\hat{\mu}$ and $\hat{\sigma}^2$). Provide your reasoning. [6]

ii) Show and verify that those estimators indeed provide a maximum. Provide your reasoning. [5]

[25]

2. a) Consider two continuous random variables Y_1 and Y_2 with joint probability density function

$$f_{Y_1,Y_2}(y_1,y_2)\begin{cases} 2y_1, & 0<y_1<1, 0<y_2<1 \\ 0, & otherwise. \end{cases}$$

i) Compute the correlation coefficient between Y_1 and Y_2, i.e. $\text{Corr}(Y_1,Y_2)$. Provide your reasoning. [3]

ii) Compute the probability $P(Y_1 \leq 0.25 | Y_2 \geq 1/3)$. Provide your reasoning. [3]

iii) Compute the variance of $3Y_1 - 2Y_2 + 5$, i.e. $\text{Var}(3Y_1 - 2Y_2 + 5)$. Provide your reasoning. [4]

iv) Are Y_1 and Y_2 uncorrelated? Independent? Provide your reasoning. [3]

b) Consider two exponentially distributed random variables with parameter $\lambda = 1$, i.e. $X \sim EXPO(1)$ and $Y \sim EXPO(1)$. X and Y are independent.

i) Compute the joint probability density function $f_{X,Y}(x,y)$. Provide your reasoning. [4]

ii) Make the change of random variable $U = XY$ and compute the probability $P(U \leq \varepsilon)$ (with $\varepsilon > 0$). Provide your reasoning.
N.B. The following relationship will be helpful: $\int_0^\infty e^{-\left(\frac{\varepsilon}{x}+x\right)} dx = 2\sqrt{\varepsilon} K_1\left(2\sqrt{\varepsilon}\right)$ with $K_1(x)$ the modified Bessel function of the second kind. [8]